Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

Gajavada Sanjeevkumar

Problem Statement

Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

Step 1: Draw the Generating Circle ($D = 50$ mm, $R = 25$ mm)

C_0	$\rightarrow$ Initial Center Point
Circle	$\rightarrow$ Generating Circle

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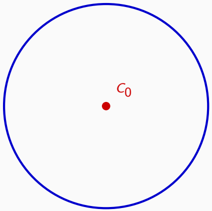

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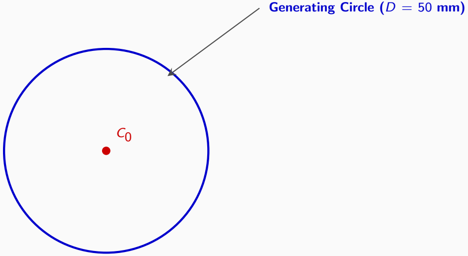

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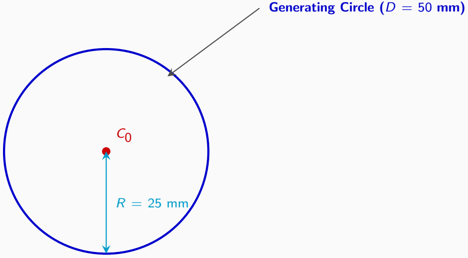

C_0 $\rightarrow$ Initial Center Point
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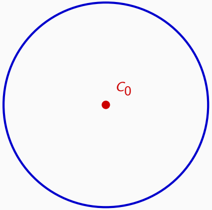
Step 2: Draw Directing Line AA' ($\pi D = 157$ mm) and Axis Line

AA'	$\rightarrow$ Directing Line (Base Line)
Axis Line	$\rightarrow$ Center Line



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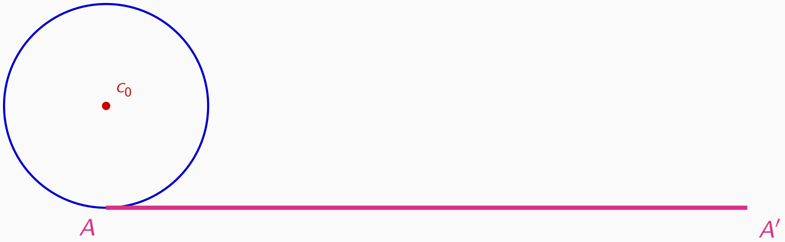

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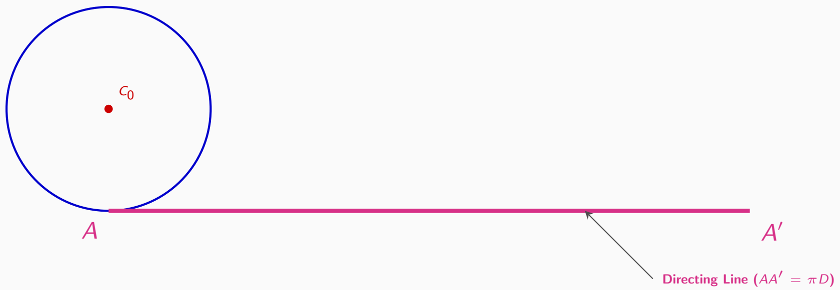

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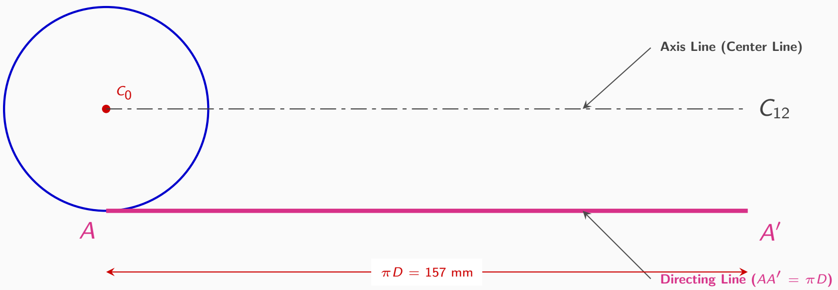

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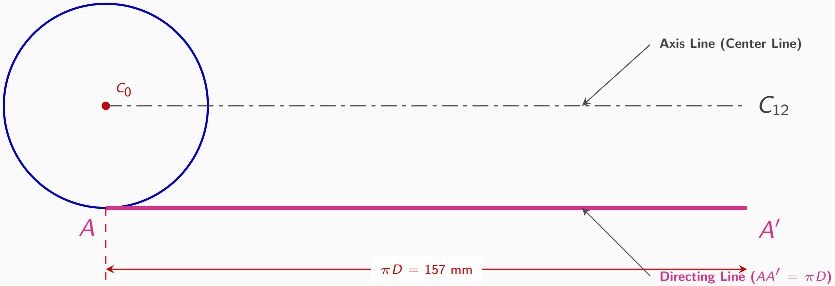

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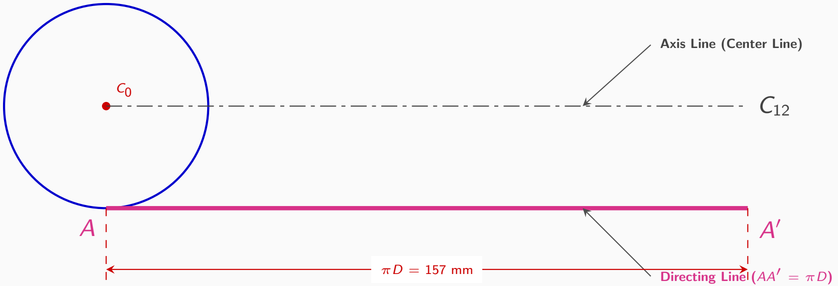

Step 3: Divide Generating Circle into 12 Equal Parts

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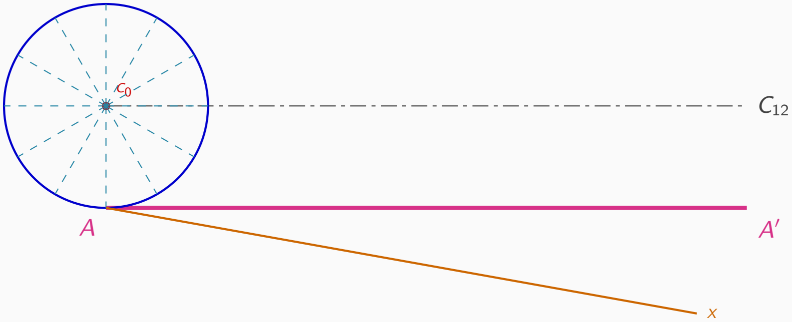

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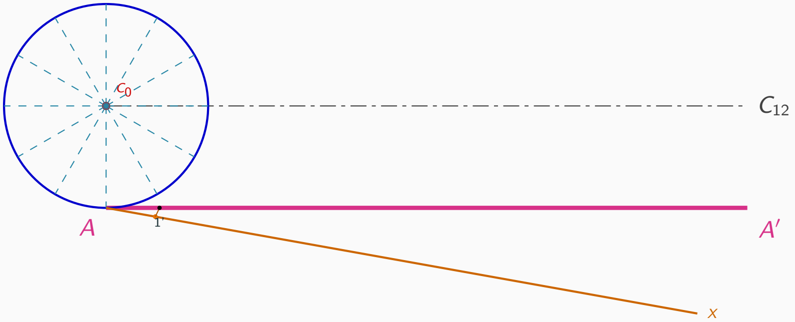

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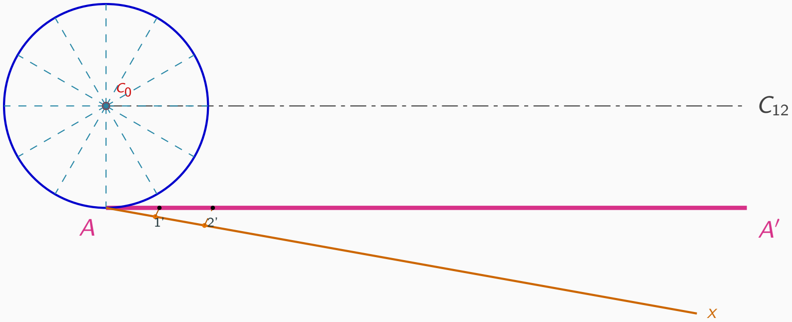

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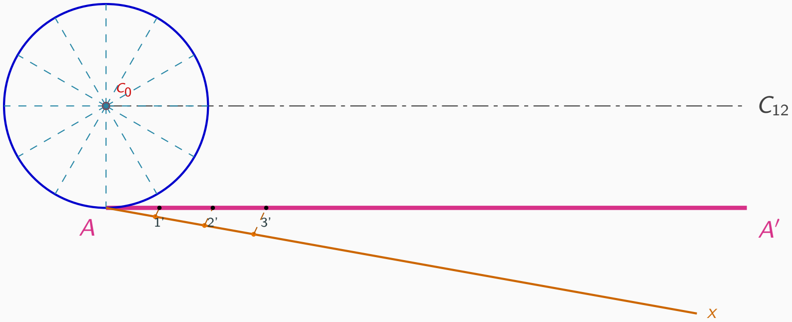

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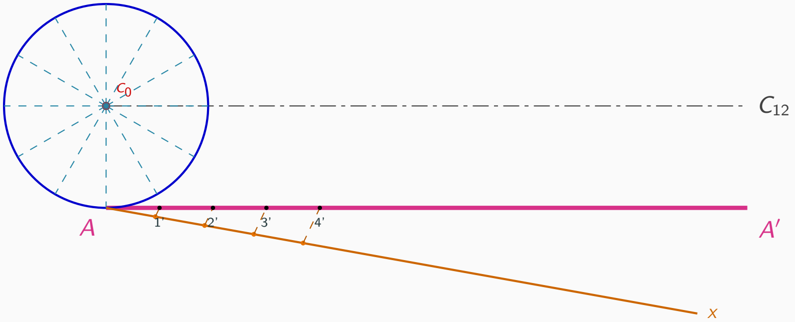

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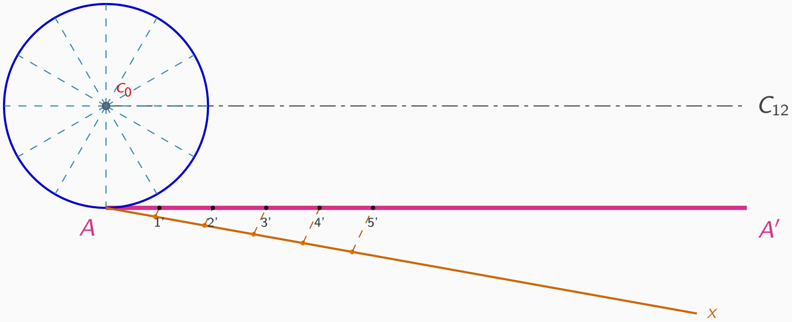

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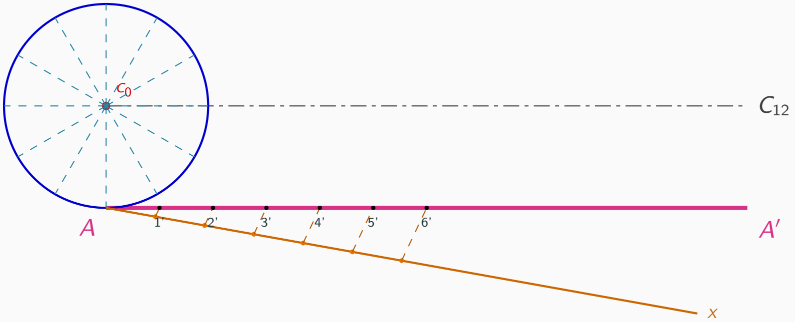

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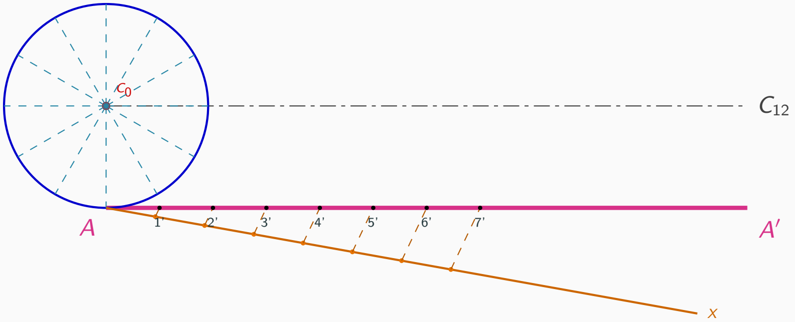

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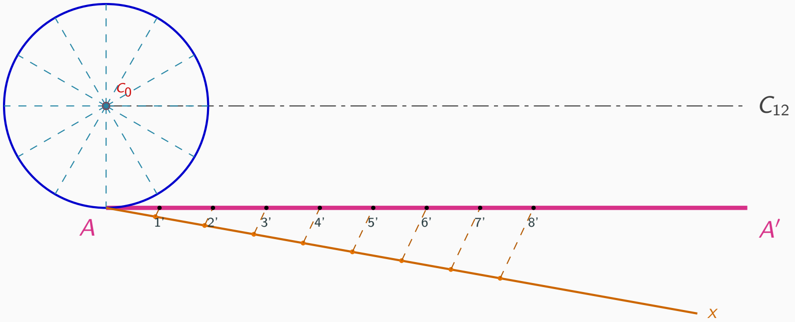

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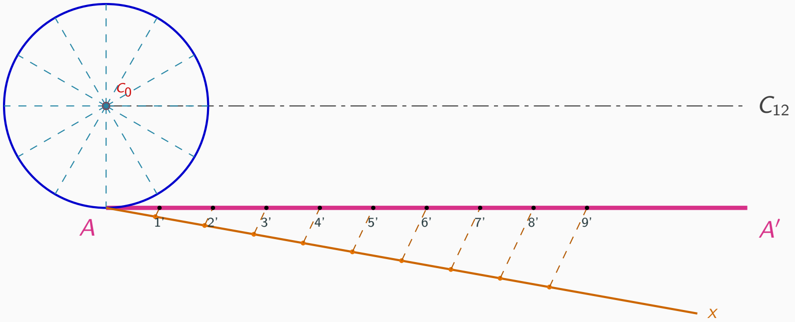

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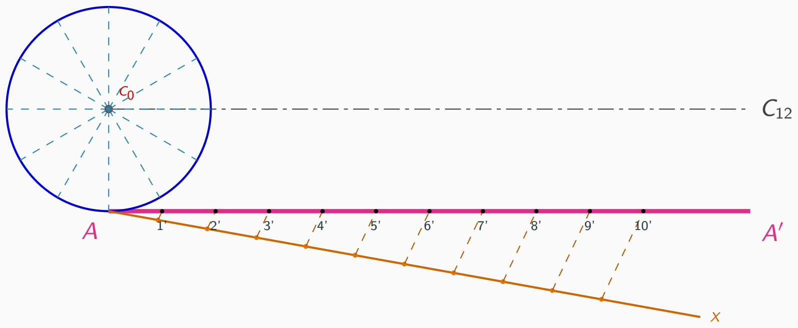

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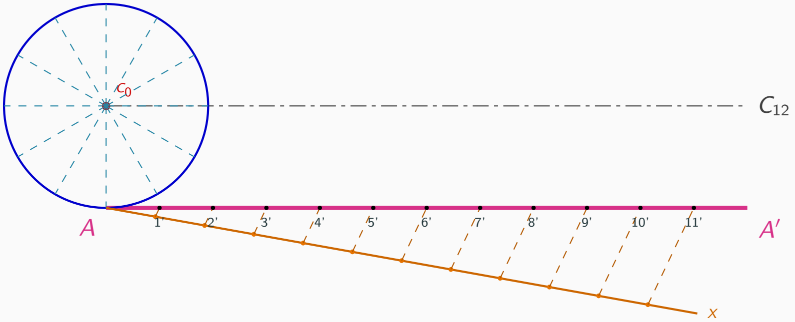

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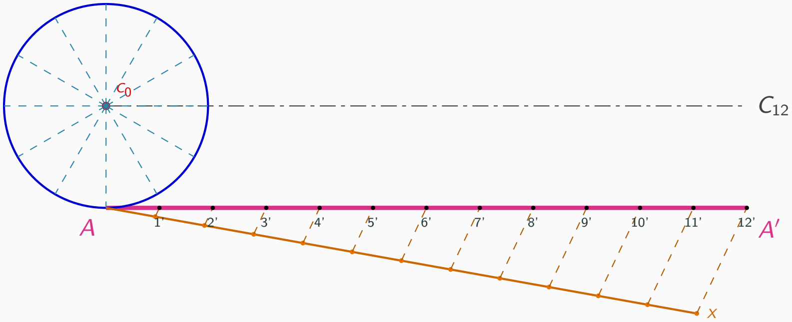

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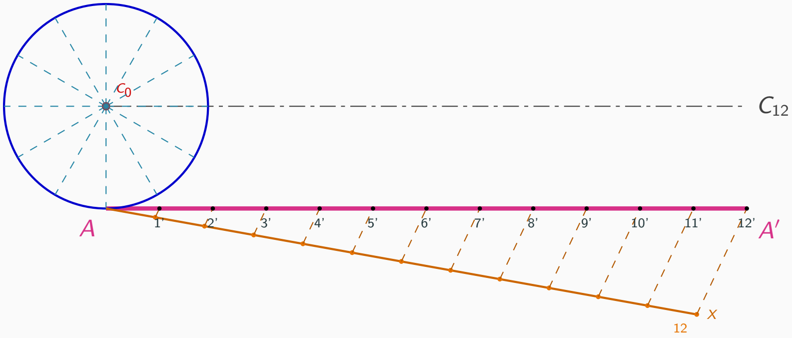

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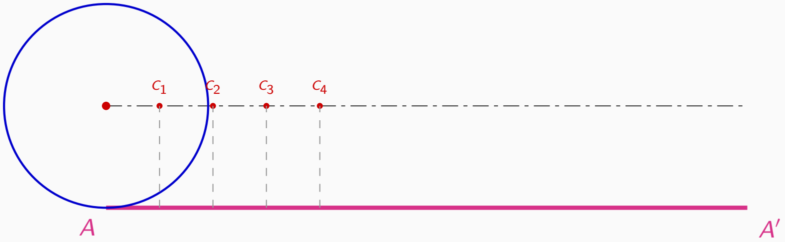

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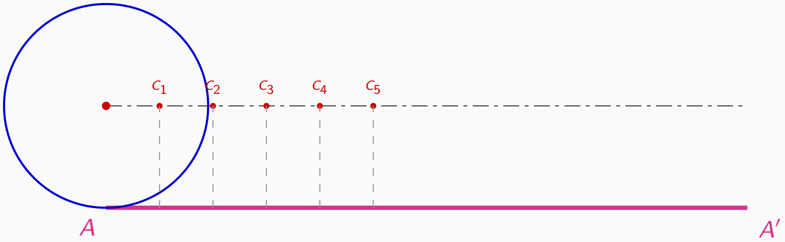

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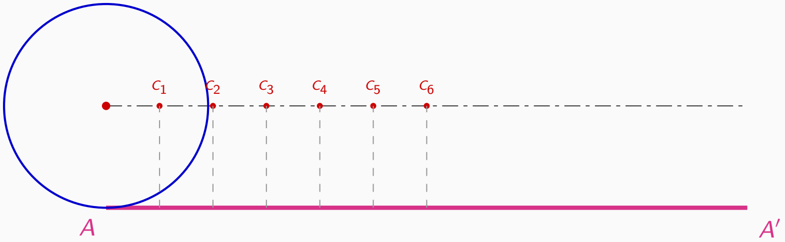

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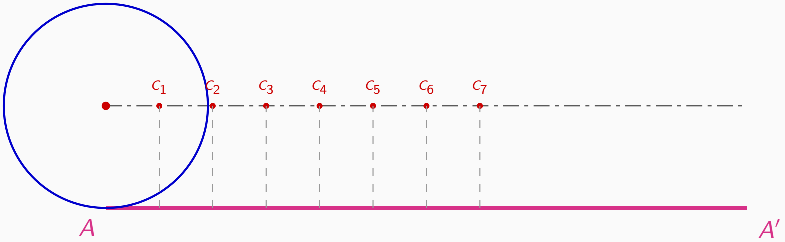

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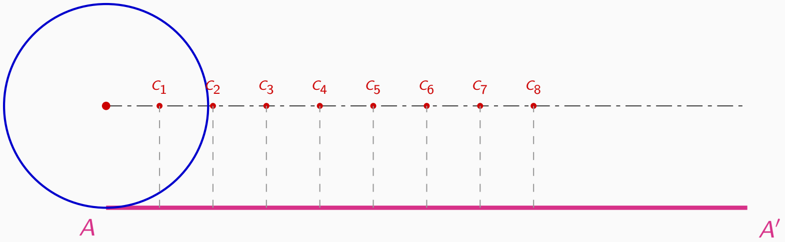

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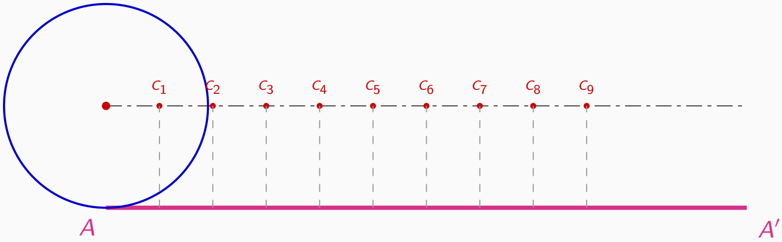

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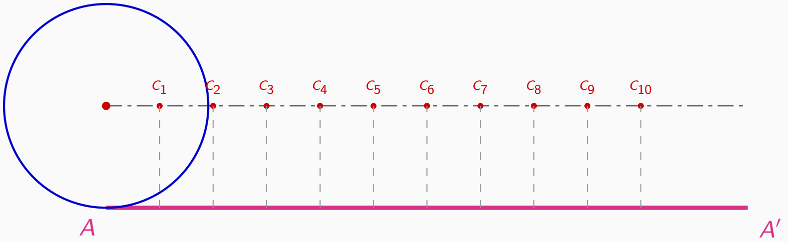

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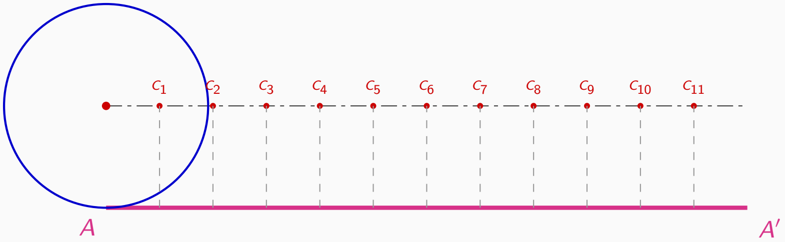

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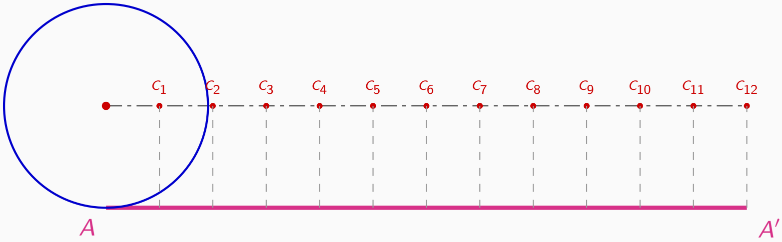

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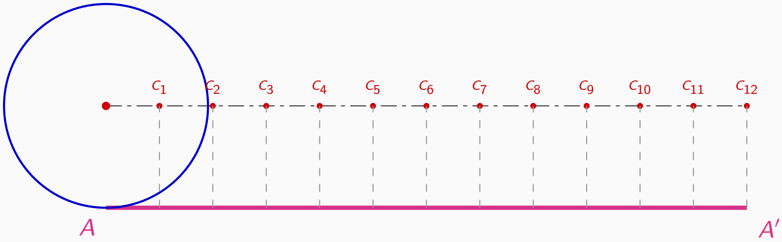
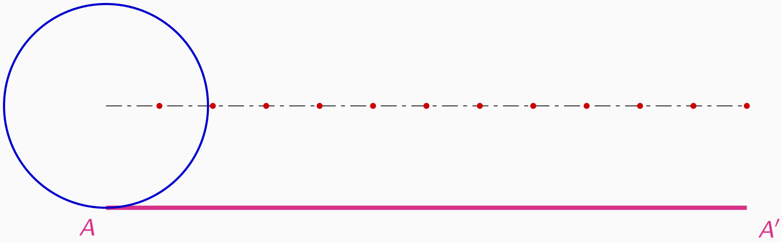


Step 6: Draw Horizontal Reference Lines

Blue Lines $\longrightarrow$ Horizontal Reference Lines
$C_1 \ldots C_{12}$ $\longrightarrow$ Retained Centers



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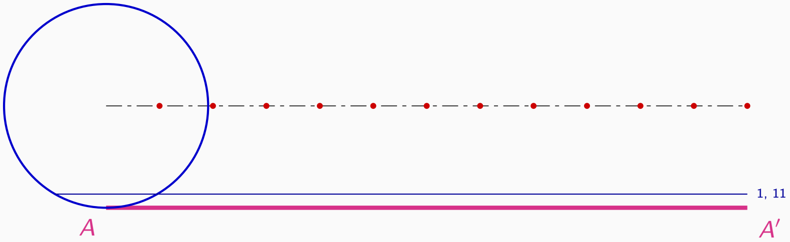

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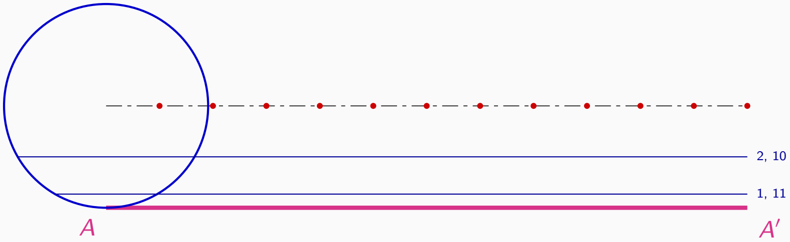

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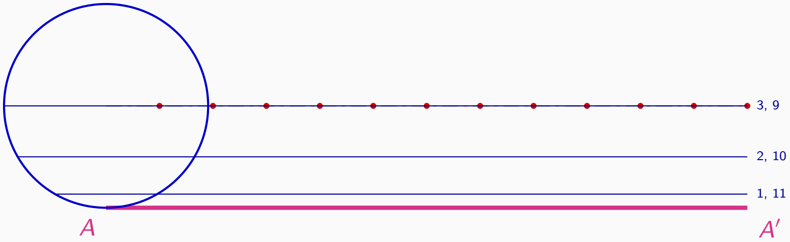

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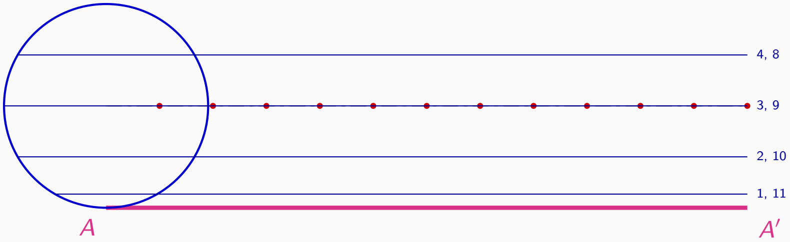

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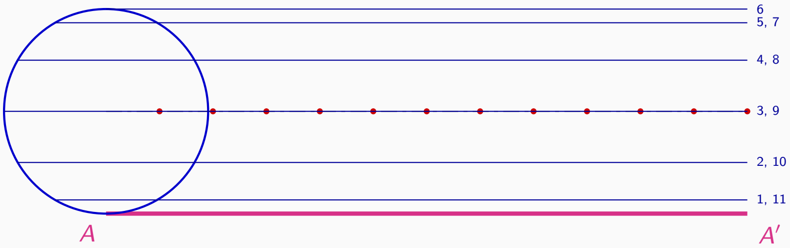
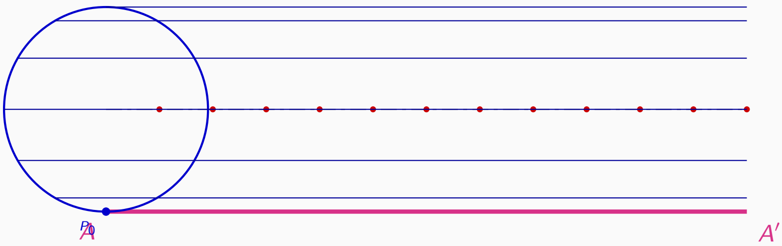


Step 7: Strike Arcs ($R = 25$ mm) to Locate Points $P_1 \ldots P_{12}$

$C_1 \ldots C_{12} \rightarrow$ Retained Centers
$P_1 \ldots P_{12} \rightarrow$ Cycloid Locus Points



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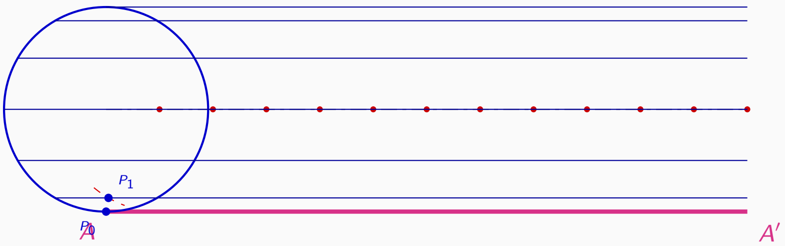

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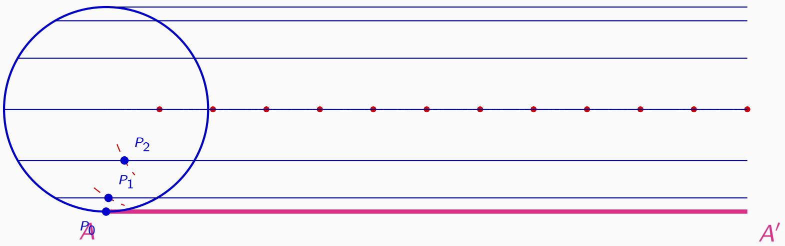

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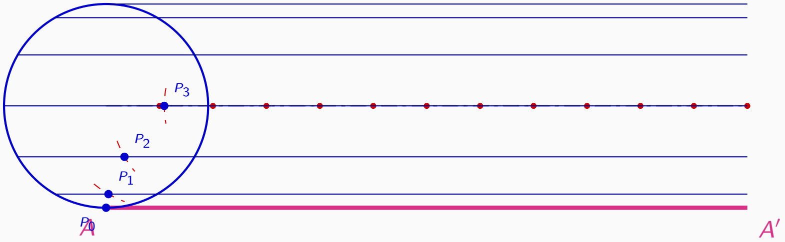

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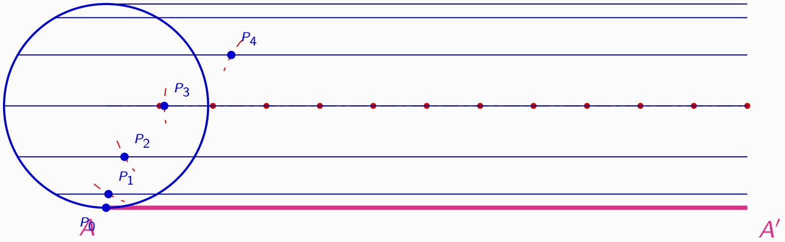

Step 7: Strike Arcs ($R = 25$ mm) to Locate Points $P_1 \dots P_{12}$

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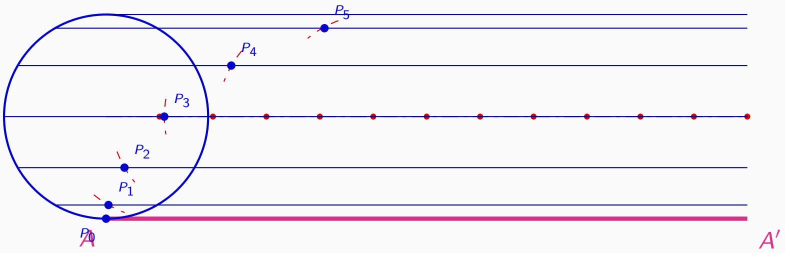

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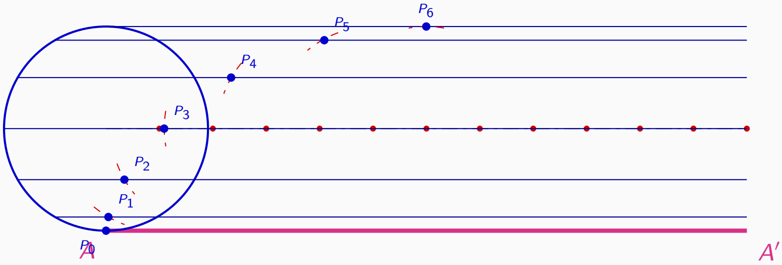

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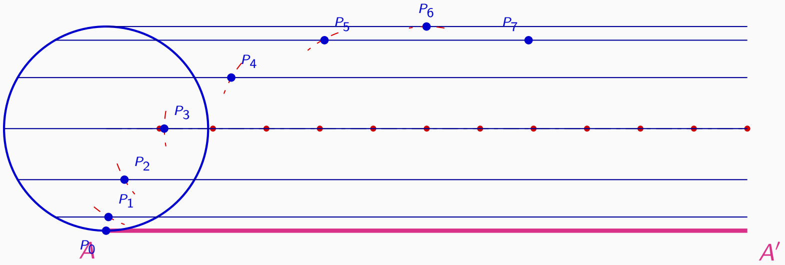

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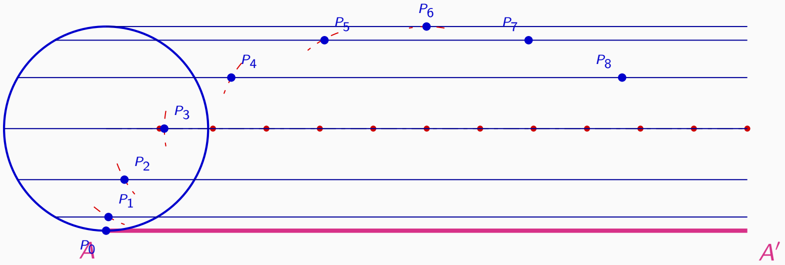

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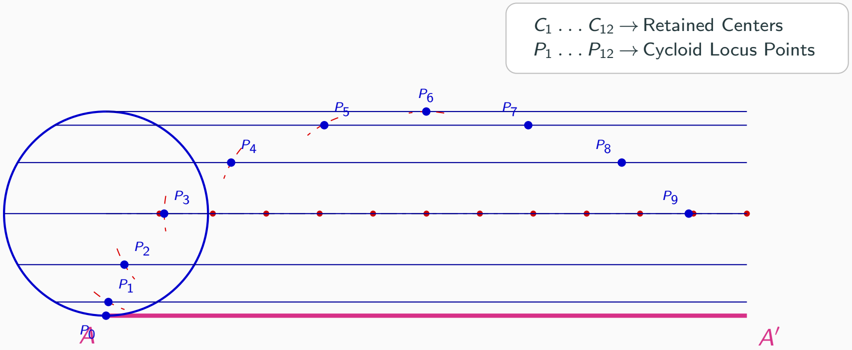

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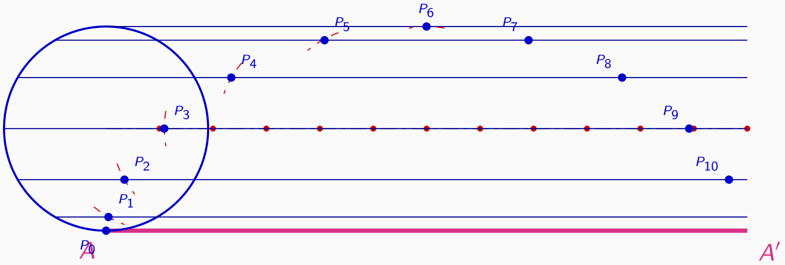

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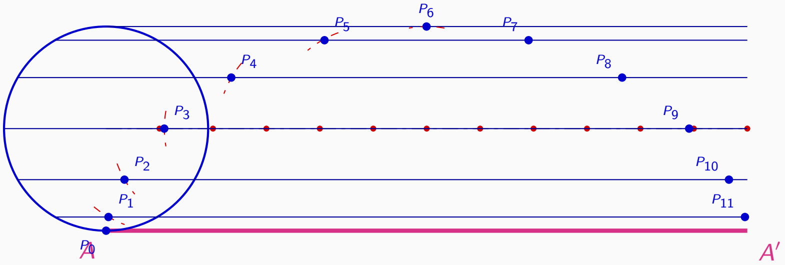

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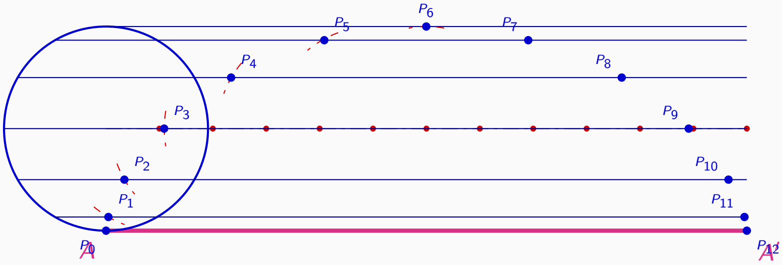
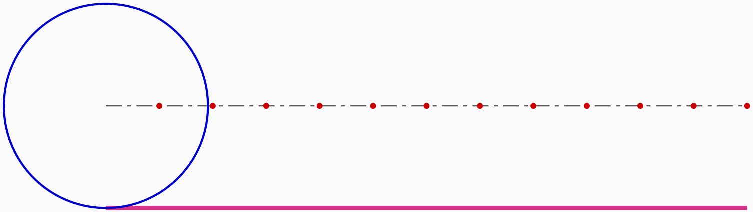


Step 8: Smoothly Join Points $P_0 \ldots P_{12}$



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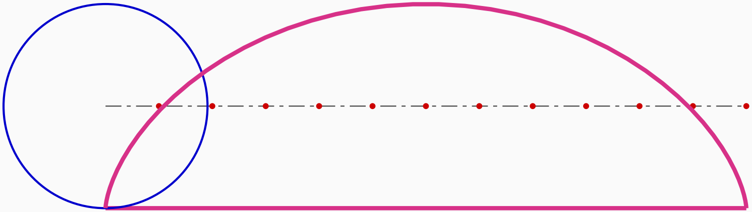
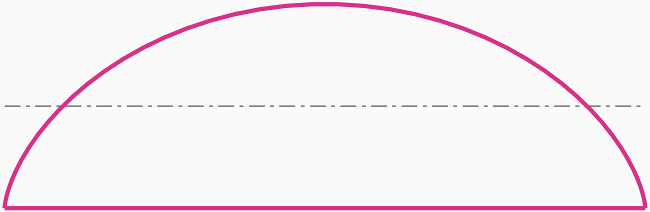


Step 9: Mark Point P & Strike Arc to Axis Line



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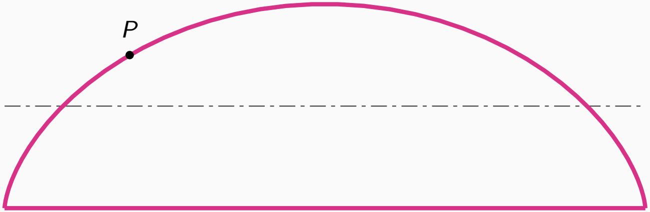

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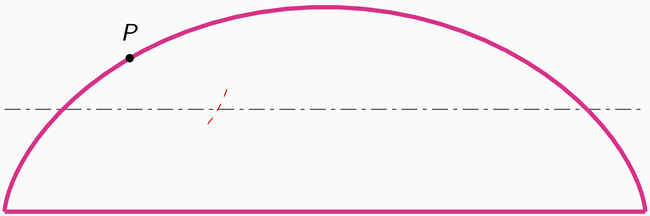

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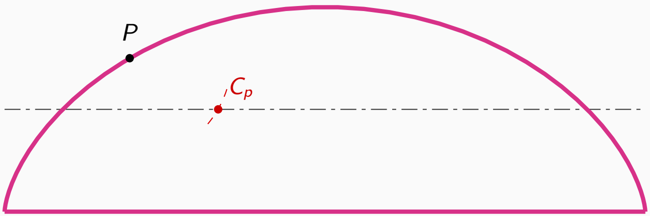

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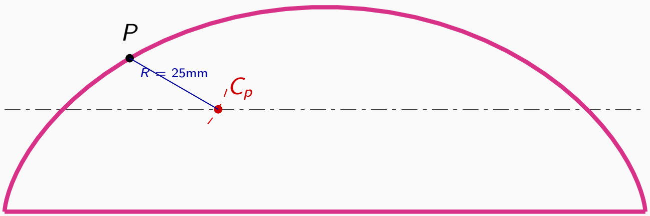
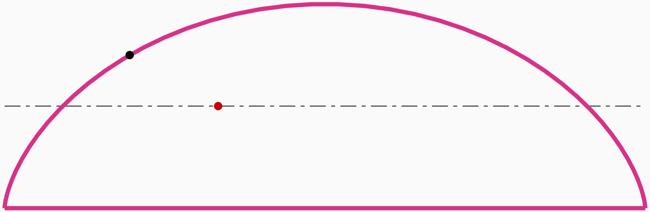


Step 10: Project C_p Vertically to Locate Normal N-N'



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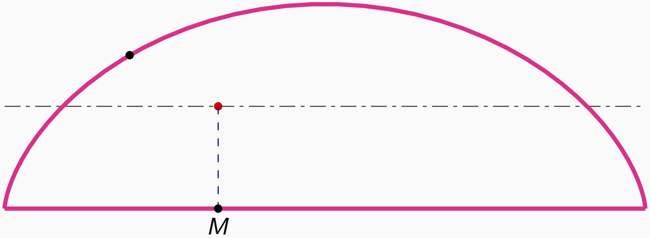

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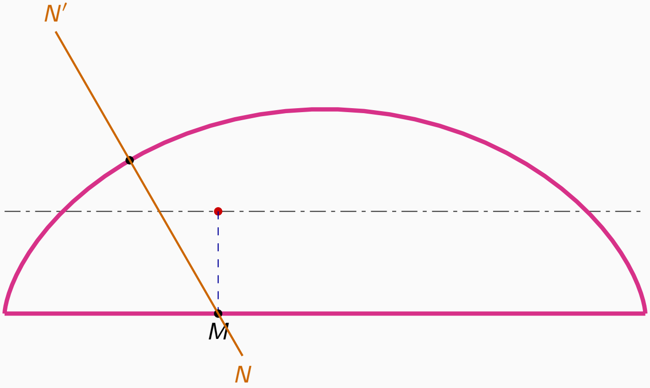
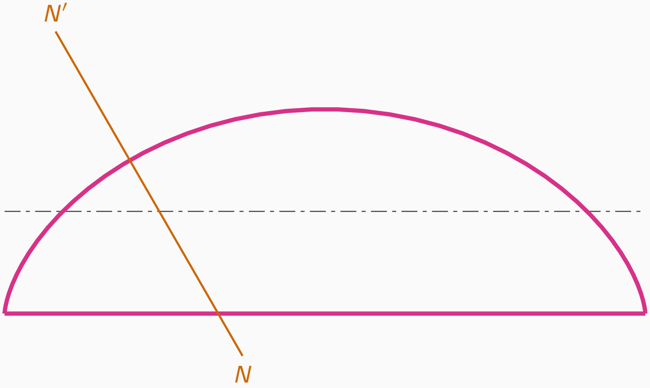


Step 11: Draw Tangent Line T-T' Perpendicular to Normal



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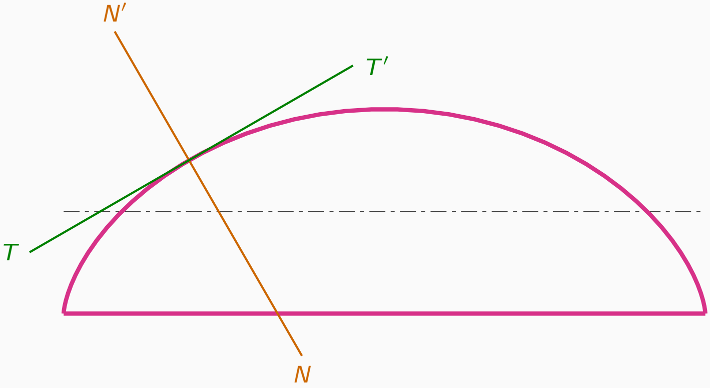

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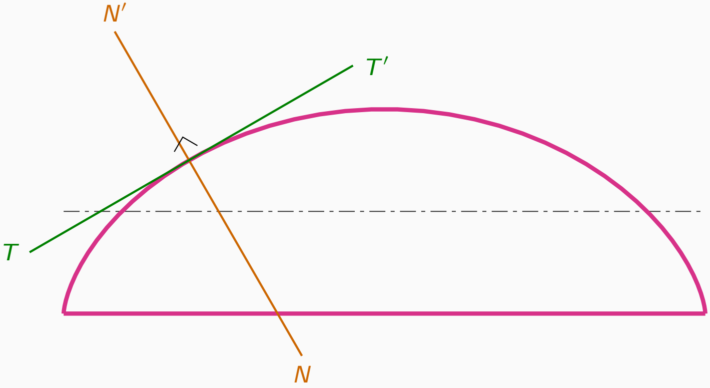
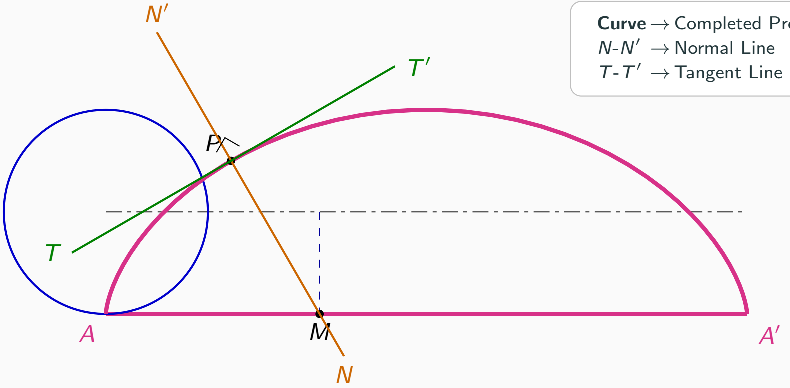
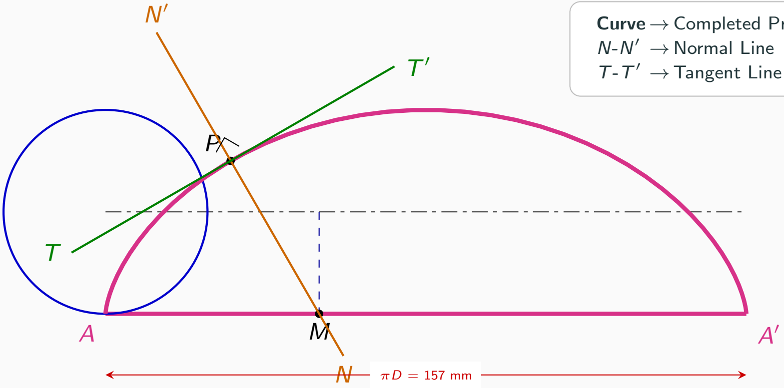


Step 12: Final Completed Cycloid with Tangent & Normal



Curve $\longrightarrow$ Completed Profile
N-N' $\longrightarrow$ Normal Line
T-T' $\longrightarrow$ Tangent Line

Step 12: Final Completed Cycloid with Tangent & Normal



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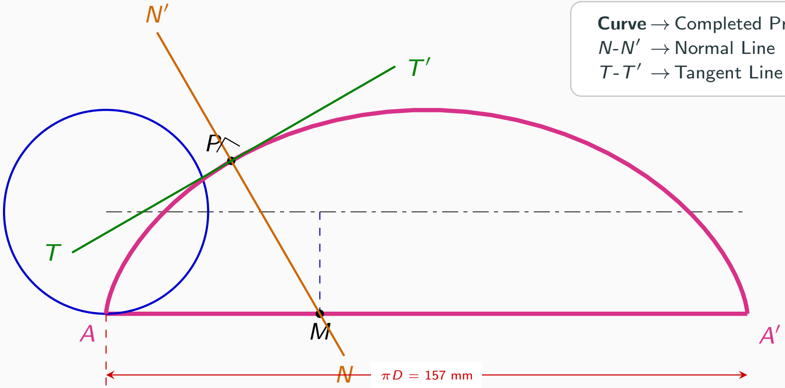

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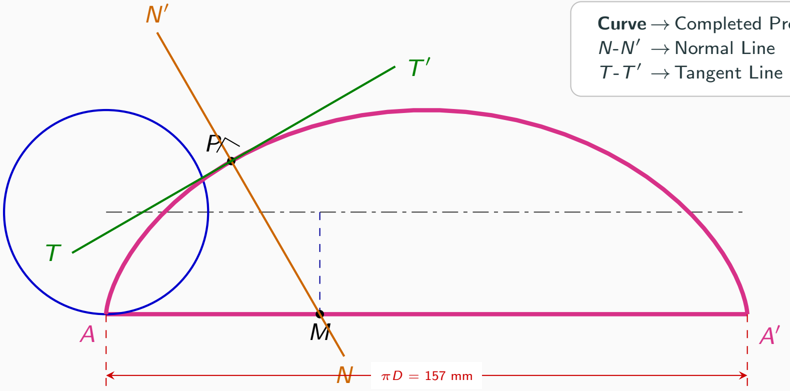

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